

Periodic Motion

→ The Motion that repeats in equal interval of time is called periodic motion.

e.g.: Circular motion, rotational motion, simple harmonic motion etc.

• Simple harmonic motion (SHM)

→ The back and forth motion of a particle in which restoring force acting on the particle is directly proportional to its displacement from the mean position and the restoring force is directed towards the mean position is called Simple harmonic motion.

So, For S.H.M.

Restoring force $\propto$ Displacement

or, $F \propto y$

$$\therefore F = -ky \rightarrow (1)$$

Where,

k is force constant and the negative sign indicates that the restoring force is directed towards the mean position.

If 'm' be the mass and 'a' be the acceleration of the particle then eqn (1) can be written as

$$ma = -ky$$

$$\Rightarrow a = -\frac{k}{m}y \rightarrow (2)$$

$$\Rightarrow a \propto -y$$

Hence,

SHM is also defined as the back and forth motion of a particle in which acceleration of the particle is directly proportional to its displacement from the mean position and is directed towards the mean position.

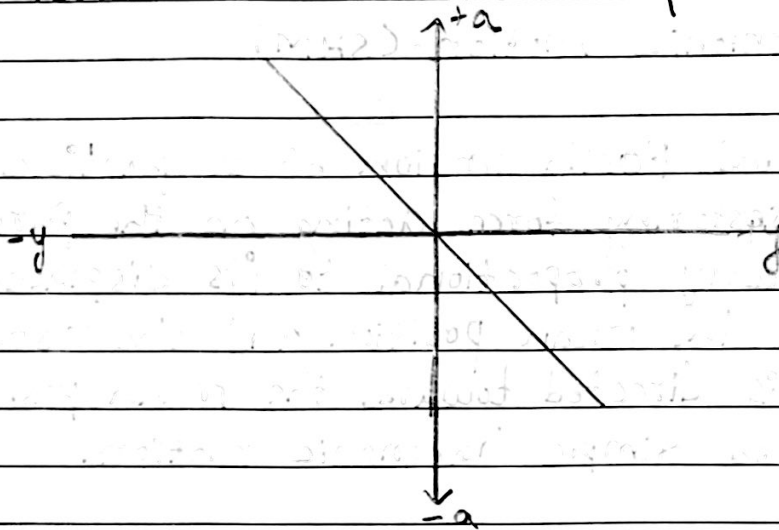
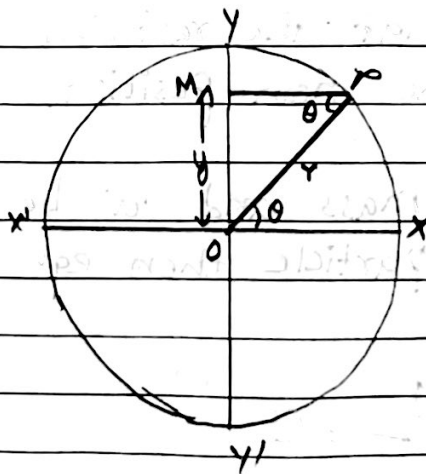


fig: Graph between acceleration and displacement in SHM

• Equation of SHM



Consider a particle is moving in a circular path of radius (r) with uniform angular velocity (ω). Let ' θ ' be the angular displacement of the particle when it moves from x to p . in time ' t '.

~~As the particle move~~

Let us draw $PM \perp YY'$. As the particle moves in circular path then the projection ' M ' oscillates about YY' with SHM. So, In the figure y represents displacement of SHM.

① Displacement (y)

→ From figure, In $\triangle OPM$,

$$\sin \theta = \frac{OM}{OP}$$

$$\Rightarrow \sin \theta = \frac{y}{r}$$

$$\Rightarrow y = r \sin \theta \rightarrow \text{①}$$

Since,

$$\theta = \omega t$$

$$\therefore y = r \sin \omega t \rightarrow \text{②}$$

Case-I: When $\theta = \omega t = 0^\circ$ then $y = 0$ (mean position)

Case-II: When $\theta = \omega t = 90^\circ$ then $y = r$ (Extreme position)

Here, r is the maximum value of displacement which is called Amplitude.

• Amplitude

→ The maximum displacement is called amplitude.

② Velocity (v)

→ It is the rate of change of displacement with time.

$$\therefore v = \frac{dy}{dt}$$

$$= \frac{d(r \sin \omega t)}{dt}$$

$$= r \cos \omega t \cdot \omega$$

$$= r \omega \cos \omega t \rightarrow \textcircled{3}$$

$$= r \omega \sqrt{1 - \sin^2 \omega t}$$

$$= r \omega \sqrt{1 - \frac{y^2}{r^2}} \quad (\text{from } \textcircled{2})$$

$$= r \omega \sqrt{\frac{r^2 - y^2}{r^2}}$$

$$= \frac{r \omega}{r} \times \sqrt{r^2 - y^2}$$

$$= \omega \sqrt{r^2 - y^2} \rightarrow \textcircled{4}$$

Case-I : At mean position i.e. $y=0$

$$v = \omega r \quad (\text{maximum})$$

$$\therefore v_{\text{max}} = \omega r$$

Case-II : At extreme position, $y=r$ then

$$v=0$$

③ Acceleration

→ The rate of change of velocity with time is

called acceleration.

$$\therefore a = \frac{dv}{dt}$$

$$a = \frac{d(r\omega \cos \omega t)}{dt} \quad [\text{from (3)}]$$

$$= r\omega (-\sin \omega t) \cdot \omega$$

$$= -\omega^2 r \sin \omega t$$

$$= -\omega^2 y \rightarrow (5)$$

Case-I: At mean position $y=0$, then
 $a=0$

Case-II: At extreme position $y=r$.
 $a = -\omega^2 r$

$$\therefore \underline{a = +\omega^2 r} \quad (\text{In magnitude})$$

$\hookrightarrow$ Max. value.

$$\therefore \boxed{a_{\text{max}} = \omega^2 r}$$

(v) Time period (T)

$\hookrightarrow$ The time taken by a particle to complete one oscillation is called time period.

From eqn (5)

$$a = -\omega^2 y$$

$$\Rightarrow a = \omega^2 y \quad (\text{In magnitude})$$

$$\Rightarrow \omega^2 = \frac{a}{y}$$

$$\Rightarrow \omega = \sqrt{\frac{a}{y}}$$

$$\Rightarrow \frac{2\pi}{T} = \sqrt{\frac{a}{y}}$$

$$\therefore T = 2\pi \sqrt{\frac{y}{a}} \rightarrow 2\pi \sqrt{\frac{\text{displacement}}{\text{Acceleration}}} \rightarrow \textcircled{6}$$

⑤ Frequency (f)

→ Number of oscillations per second is called frequency.

It is given by reciprocal of time period.
i.e. $f = \frac{1}{T}$

$$\therefore f = \frac{1}{2\pi} \sqrt{\frac{a}{y}} \rightarrow \textcircled{7}$$

⑥ Phase

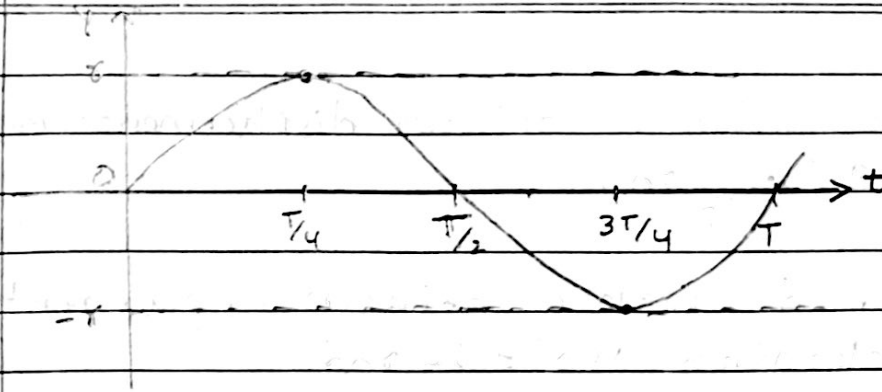
→ The phase of a particle executing S.H.M is the state of motion of particle regarding its position and direction of motion. It is measured in terms of angle. Eg: In the equation $y = r \sin \omega t$, ωt is the phase angle.

• Graph of displacement, velocity and acceleration in S.H.M

① Displacement

$$y = r \sin \omega t = r \sin \frac{2\pi}{T} t$$

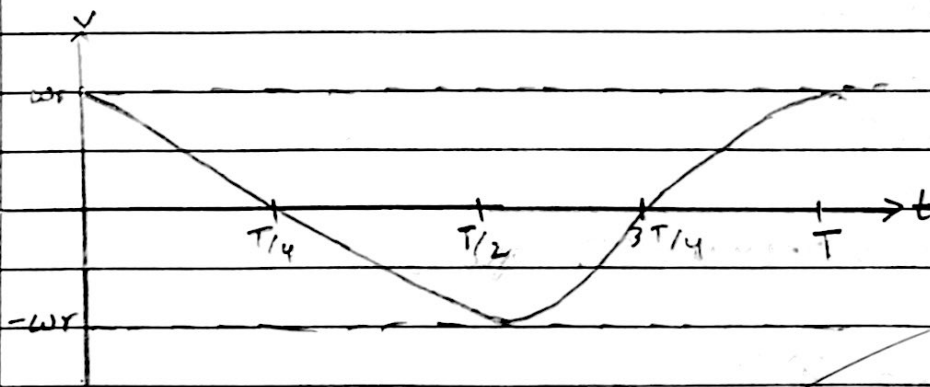
t	0	T/4	T/2	3T/4	T
y	0	r	0	-r	0



② Velocity

$$v = \omega r \cos \omega t = \omega r \cos \frac{2\pi}{T} t$$

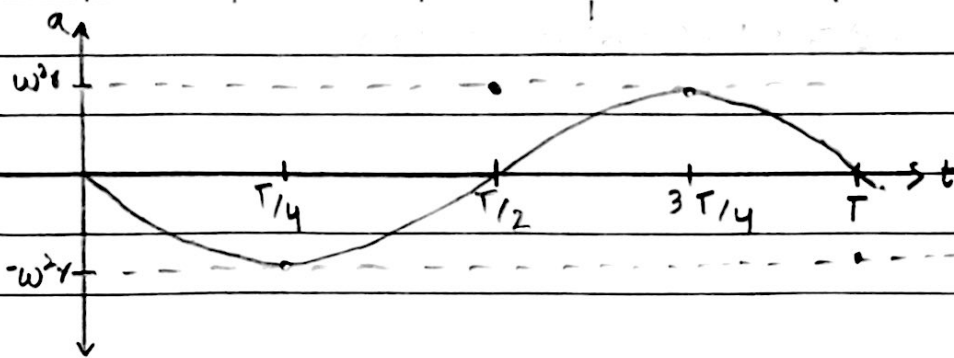
t	0	$T/4$	$T/2$	$3T/4$	T
v	ωr	0	$-\omega r$	0	ωr



③ Acceleration

$$a = -\omega^2 y = -\omega^2 r \sin \omega t = -\omega^2 r \sin \frac{2\pi}{T} t$$

t	0	$T/4$	$T/2$	$3T/4$	T
a	0	$-\omega^2 r$	0	$\omega^2 r$	0



Note:

- ① The phase difference between displacement and velocity $= 90^\circ = \frac{\pi}{2}$ rad
- ② The phase difference between displacement and acceleration $= 180^\circ = \pi$ rad.

① Qn A body is vibrating with S.H.M of amplitude 15 cm and frequency of 4 Hz. Calculate the max. value of acceleration and velocity.

→ Soln,

$$x = 15 \text{ cm} = 15 \times 10^{-2} \text{ m}$$

$$f = 4 \text{ Hz}$$

$$a_{\text{max}} = ?$$

$$v_{\text{max}} = ?$$

Now,

$$\omega = 2\pi f = 2\pi \times 4 = 8\pi \text{ rad/s}$$

Then,

$$a_{\text{max}} = \omega^2 x$$

$$= (8\pi)^2 \times 15 \times 10^{-2}$$

$$= 64 \times (3.14)^2 \times 15 \times 10^{-2}$$

$$= ~~9.48 \text{ m/s}^2~~ 94.8 \text{ m/s}^2$$

And

$$v_{\text{max}} = \omega x$$

$$= 8\pi \times 15 \times 10^{-2}$$

$$= ~~9.48 \text{ m/s}^2~~ 3.77 \text{ m/s}$$

- ② A body of mass 200 gm is executing simple H.M with amplitude of 20 mm. The max. force which acts upon it is 0.8 N. calculate its max. velocity and its period of oscillation.

→ Soln.

$$m = 200 \text{ gm} = \frac{200}{1000} = 0.2 \text{ kg}$$

$$r = \frac{20}{1000} \text{ m} = 0.02 \text{ m}$$

$$F_{\text{max}} = 0.8 \text{ N}$$

$$V_{\text{max}} = ?$$

$$T = ?$$

We know that

$$F_{\text{max}} = m a_{\text{max}}$$

$$0.8 = 0.2 \times \omega^2 r$$

$$\Rightarrow 4 = \omega^2 \times 0.02$$

$$\Rightarrow \omega^2 = 200$$

$$\therefore \omega = 10\sqrt{2} \text{ rad/s}$$

Then,

$$V_{\text{max}} = \omega r$$

$$= 10\sqrt{2} \times 0.02$$

$$= 0.28 \text{ m/s}$$

and,

$$\omega = \frac{2\pi}{T}$$

$$T = \frac{2\pi}{10\sqrt{2}}$$

$$= 0.45 \text{ s}$$

- ③ The displacement of a particle executing S.H.M is given by $y = 0.01 \sin 50\pi t$ where y is in meter and t in sec.

Find ① Amplitude, frequency and time period of particle.

- ② Velocity of particle at $t = 0$ s.

→ Soln,

$$y = 0.01 \sin 50\pi t$$

- ① $x = ?$, $f = ?$ and $T = ?$

Comparing given eqⁿ with $y = x \sin \omega t$, we have.

$$x = 0.01 \text{ m and}$$

$$\omega = 50\pi$$

$$\text{or, } 2\pi f = 50\pi$$

$$\therefore f = 25 \text{ Hz.}$$

Now,

$$T = \frac{1}{f} = \frac{1}{25} = 0.04 \text{ Sec.}$$

- ② $V = ?$ when $t = 0$

When $t = 0$ then, $y = 0.01 \sin 50\pi t$

$$y = 0.01 \times \sin 0$$

$$y = 0.01 \times 0$$

$$= 0$$

Then,

$$V = \omega \sqrt{x^2 - y^2}$$

$$= 50\pi \sqrt{(0.01)^2 - 0}$$

$$= 50\pi \times 0.01$$

$$= 1.57 \text{ m/s.}$$

- ⑨ A small mass rests on a horizontal platform which vibrates in S.H.M with a period of 0.25 s. Find the maximum amplitude of the motion which will allow mass to remain in contact with the platform through out the motion.

→ Soln,

$$T = 0.25 \text{ sec}$$

$$r = ?$$

Here,

For the mass to remain in contact with the platform, the maximum acceleration of the platform is equal to g .

$$\text{i.e. } a_{\text{max}} = g$$

$$\text{or, } \omega^2 r = g$$

$$\Rightarrow \left(\frac{2\pi}{T} \right)^2 \times r = g$$

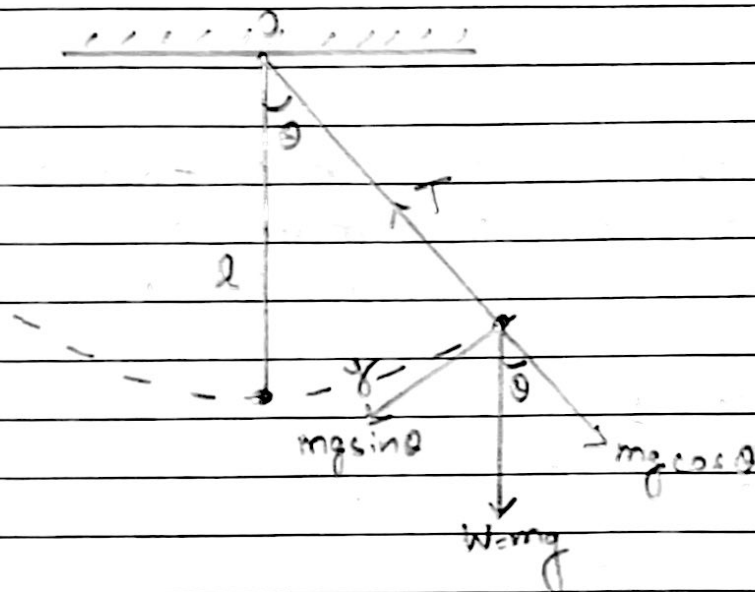
$$\text{or, } r = \frac{g T^2}{4\pi^2}$$

$$\text{or, } r = \frac{9.8 \times (0.25)^2}{4 \times (3.14)^2}$$

$$\text{or, } r = 0.015 \text{ m}$$

- Simple pendulum

→ A simple pendulum is a heavy particle (or bob) suspended from a rigid support by an inextensible and weightless string which is free to oscillate in a vertical plane.



Let θ be the angular displacement and y be the linear displacement of pendulum when it moves from mean position A to any position B. Let l be the length and ' m ' be the ~~acete~~ mass of the pendulum.

The different forces acting on the pendulum are:

- (i) Its weight, $W = mg$
- (ii) Tension, T .

Here, the weight mg is resolved into two rectangular components $mg \sin \theta$ and $mg \cos \theta$

as shown in figure.

The component $mg \sin \theta$ acts towards the mean position and provides restoring force to the pendulum.

$$\therefore \text{Restoring force (F)} = -mg \sin \theta$$

$$\text{or, } ma = -mg \sin \theta$$

$$\therefore a = -g \sin \theta$$

$$\text{For small } \theta, \sin \theta \approx \theta = \frac{y}{l}$$

$$\therefore a = -\frac{g}{l} y \rightarrow \textcircled{1}$$

Here, $\frac{g}{l}$ is constant

$$\therefore a \propto -y$$

This proves that the motion of simple pendulum is S.H.M.

For S.H.M,

$$\text{we have, } a = -\omega^2 y \rightarrow \textcircled{2}$$

From ① & ②

$$-\frac{g}{l} y = -\omega^2 y$$

$$\Rightarrow \omega^2 = \frac{g}{l}$$

$$\omega = \sqrt{\frac{g}{l}}$$

$$\text{or, } \frac{2\pi}{T} = \sqrt{\frac{g}{l}}$$

$$\text{or, } \therefore T = 2\pi \sqrt{\frac{l}{g}} \rightarrow \textcircled{3}$$

This gives time period of simple pendulum.

And, frequency (f) = $\frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{g}{L}}$

- Second pendulum

→ The simple pendulum having time period 2 second is called second pendulum.
i.e. For second pendulum, $T = 2 \text{ sec}$.

- Motion of simple pendulum inside a lift

① When lift is at rest or moving with constant velocity (i.e. $a = 0$)

$$T = 2\pi \sqrt{\frac{L}{g}} \quad [\because g_{\text{effective}} = g]$$

② When lift is moving upward with acceleration a .

$$T = 2\pi \sqrt{\frac{L}{g+a}} \quad [\because g_{\text{effective}} = g+a]$$

③ When lift is moving downward with acceleration a .

$$T = 2\pi \sqrt{\frac{L}{g-a}} \quad [\because g_{\text{effective}} = g-a]$$

→ When lift is in free fall i.e. $a = g$ then

$$T = 2\pi \sqrt{\frac{L}{g-g}} = \infty$$

i.e. There is no oscillation of simple pendulum.

Q. Can we use pendulum clock in an artificial satellite.

→ Artificial satellites are always in free fall (i.e. $a=g$) Then, $T = 2\pi \sqrt{\frac{L}{g-g}} \Rightarrow T = \infty$

This shows that there is no oscillation of pendulum in the clock so, we cannot use pendulum clock in ~~in~~ artificial satellite.

Q.4) A simple pendulum 4 m long swings with an amplitude of 0.2 m. Compute the velocity of the pendulum at its lowest point and acceleration at its extreme end.

→ soln,

$$L = 4 \text{ m}$$

$$r = 0.2 \text{ m}$$

$$v = ? \text{ (at lowest point)}$$

$$a = ? \text{ (at extreme point)}$$

Now,

$$\omega = \sqrt{\frac{g}{L}}$$

$$= \sqrt{\frac{9.8}{4}}$$

$$= \sqrt{2.45}$$

$$= 1.56 \text{ rad/s}$$

Then,

$$v = \omega r$$

$$= 1.56 \times 0.2$$

$$= 0.31 \text{ m/s.}$$

Again,

$$a = \omega^2 r$$

$$= (1.56)^2 \times 0.2$$

$$= 0.49 \text{ m/s}^2$$

- c) A simple pendulum has a period of 4.2 Sec. When the pendulum is ~~short~~ shortened by 1m the period is 3.7 Sec. From these measurements, calculate the acceleration of freefall and the original length of pendulum.

→ Soln,

$$T_1 = 4.2 \text{ s}$$

$$\text{When } l_2 = l_1 - 1,$$

$$T_2 = 3.7 \text{ Sec}$$

So,

$$T_1 = 2\pi \sqrt{\frac{l_1}{g}}$$

$$T_2 = 2\pi \sqrt{\frac{l_2}{g}}$$

$$\Rightarrow T_2 = 2\pi \sqrt{\frac{l_1 - 1}{g}} \rightarrow (ii)$$

dividing (i) by (ii)

$$\frac{4.2}{3.7} = \sqrt{\frac{l_1}{l_1 - 1}}$$

$$\Rightarrow \frac{42}{37} = \sqrt{\frac{l_1}{l_1 - 1}}$$

$$\text{or, } \frac{1763}{1369} = \frac{l_1}{l_1 - 1}$$

$$\text{or, } 1764 L_1 - 1764 = 1369 L_1$$

$$\text{or, } 395 L_1 = 1764$$

$$\therefore L_1 = 4.46 \text{ m}$$

So,

$$g = ?$$

$$4.2 = 2\pi \sqrt{\frac{4.46}{g}}$$

$$\text{or, } (4.2)^2 = 4\pi^2 \times \frac{4.46}{g}$$

$$\text{or, } 0.4468 = \frac{4.46}{g}$$

$$\therefore g = 9.98 \text{ m/s}^2$$

⑦ A glider with mass $m = 2.00 \text{ kg}$ sits on a frictionless horizontal air track, connected to a spring with force constant $k = 5.00 \text{ N/m}$. You pull the glider, stretching the spring 0.100 m and then releases it with no initial velocity. The glider begins to move back toward its equilibrium position ($x = 0$). What is its velocity when $x = 0.080 \text{ m}$?

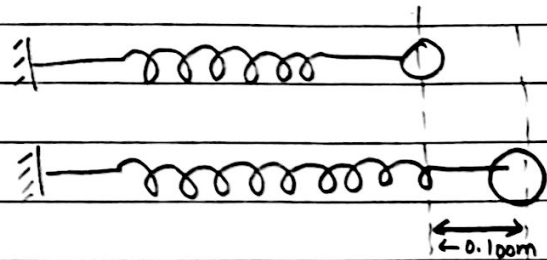
→ Soln,

$$m = 2 \text{ kg}$$

$$k = 5 \text{ N/m}$$

$$r = 0.100 \text{ m}$$

$$v = ? \text{ when } x = 0.080 \text{ m}$$



Now,

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{5}{2}} = 1.58 \text{ rad/s}$$

Then,

$$\begin{aligned}V &= \omega \sqrt{r^2 - n^2} \\&= 1.58 \sqrt{(0.100)^2 - (0.080)^2} \\&= 1.58 \times 0.06 \\&= 0.0948 \text{ m/s}\end{aligned}$$

• Vertical Mass-spring system.

Let e be the extension of spring

When a body of mass m is suspended in it. Then

From Hooke's law

$$F_1 \propto e$$

$$\therefore F_1 = -ke \rightarrow (1)$$

Where,

k is Force constant of spring.

Suppose, the ~~mass~~ body is pulled down

and released. Then

the body oscillates up and down about the

mean position O . Let y be the displacement of the body from its mean position then again from Hooke's law.

$$F_2 \propto (y+e)$$

$$\therefore F_2 = -k(y+e) \rightarrow (2)$$

Now,

Net force that causes only oscillation of body is given by

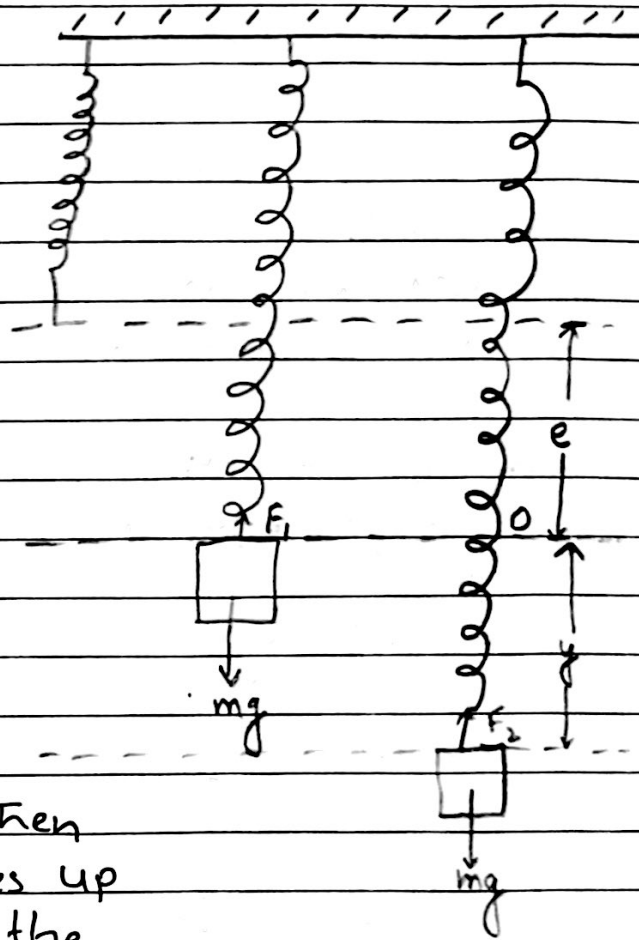
$$F_{net} = F_2 - mg$$

$$\text{or, } F_{net} = F_2 - F_1 \quad [\because F_1 = mg]$$

$$\text{or, } ma = -k(y+e) - (-ke)$$

$$= -ky - ke + ke$$

$$\text{or, } ma = -ky$$



$$\text{or, } a = -\frac{k}{m} y \rightarrow (3)$$

$$\Rightarrow a \propto -y$$

This proves that the motion of body is SHM.

For SHM, we have

$$a = -\omega^2 y \rightarrow (4)$$

From (3) & (4)

$$-\omega^2 y = -\frac{k}{m} y$$

$$\Rightarrow \omega^2 = \frac{k}{m}$$

$$\Rightarrow \omega = \sqrt{\frac{k}{m}}$$

$$\text{or, } \frac{2\pi}{T} = \sqrt{\frac{k}{m}}$$

$$\therefore T = 2\pi \sqrt{\frac{m}{k}} \rightarrow (5)$$

This is time period of vertical mass-spring system.

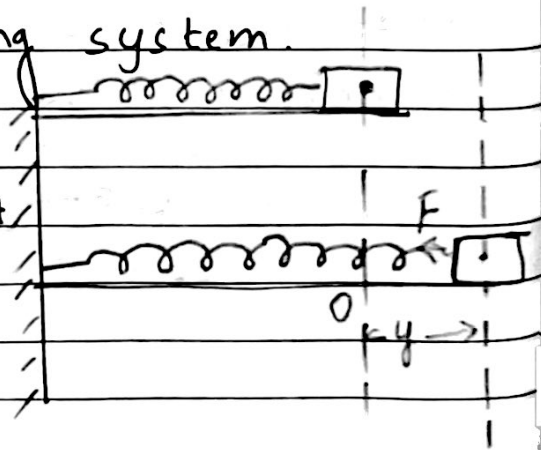
• Horizontal mass-spring system.

→ From hooke's law

Restoring force $\propto$ displacement

i.e. $F \propto y$

$$\therefore F = -ky \rightarrow (1)$$



$$\text{or, } ma = -ky$$

$$\therefore a = -\frac{k}{m}y \rightarrow (2)$$

$\Rightarrow a \propto -y$
i.e. The motion is S.H.M

For S.H.M. we have,

$$a = -\omega^2 y \rightarrow (3)$$

From (2) & (3), we get.

$$-\omega^2 y = -\frac{k}{m}y$$

$$\Rightarrow \omega = \sqrt{\frac{k}{m}}$$

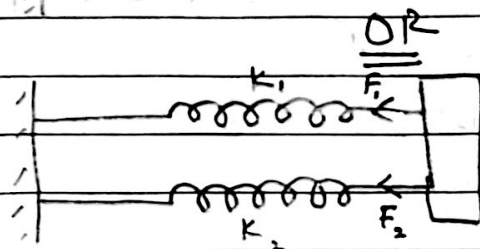
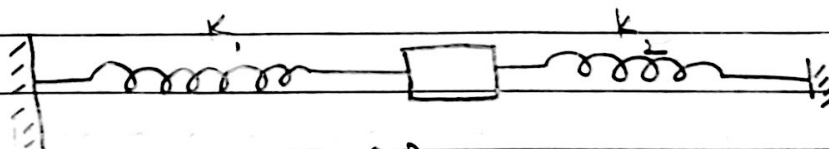
$$\Rightarrow \frac{2\pi}{T} = \sqrt{\frac{k}{m}}$$

$$\therefore T = 2\pi \sqrt{\frac{m}{k}} \rightarrow (4)$$

• Combination of springs

① Parallel combination

$$K_{eq} = K_1 + K_2$$



Proof

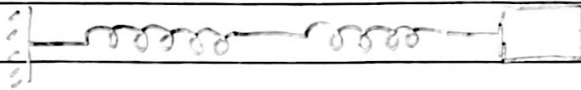
$$F_{eq} = F_1 + F_2$$

$$-K_{eq}e = -K_1e - K_2e$$

$$\boxed{K_{eq} = K_1 + K_2}$$

② Series combination.

$$\frac{1}{K_{eq}} = \frac{1}{K_1} + \frac{1}{K_2}$$



Proof : $e = e_1 + e_2$

$$\text{or, } -\frac{F}{K} = -\frac{F}{K_1} - \frac{F}{K_2}$$

$$\therefore \frac{1}{K} = \frac{1}{K_1} + \frac{1}{K_2}$$

• Energy in S.H.M

↳ A particle executing S.H.M possesses both potential energy and kinetic energy.

Potential energy

↳ The restoring force acting on a particle executing S.H.M is given by

$$F = ma = -m\omega^2 y \rightarrow \text{①}$$

$y \rightarrow \text{displacement}$

Let dy be the small displacement of particle against this force, then small workdone by the particle is given by

$$dW = -F dy$$
$$= m\omega^2 y dy \rightarrow (2)$$

Then, total workdone is given by

$$W = m\omega^2 \int_0^y y dy$$

$$= m\omega^2 \left[\frac{y^2}{2} \right]_0^y$$

$$= \frac{1}{2} m\omega^2 [y^2]_0^y$$

$$= \frac{1}{2} m\omega^2 (y^2 - 0^2)$$

$$= \frac{1}{2} m\omega^2 y^2 \rightarrow (3)$$

This workdone stores in the particle as its potential energy.

$$\text{i.e. P.E} = W$$

$$\therefore \text{P.E} = \frac{1}{2} m\omega^2 y^2$$

Kinetic Energy

→ The K.E of particle is given by

$$\text{K.E} = \frac{1}{2} mv^2$$

$$= \frac{1}{2} m (\omega \sqrt{r^2 - y^2})^2$$

$$K.E = \frac{1}{2} m \omega^2 (r^2 - y^2) \rightarrow (5)$$

> Total energy

↳ It is the sum of potential energy and K.E.

$$\therefore T.E = P.E + K.E$$

$$= \frac{1}{2} m \omega^2 y^2 + \frac{1}{2} m \omega^2 (r^2 - y^2)$$

$$= \frac{1}{2} m \omega^2 (y^2 + r^2 - y^2)$$

$$= \frac{1}{2} m \omega^2 r^2$$

$$= \text{constant}$$

i.e. total energy of particle in SHM remains constant.

• Graph of P.E and K.E with displacement in SHM

↳ Since,

$$P.E = \frac{1}{2} m \omega^2 y^2$$

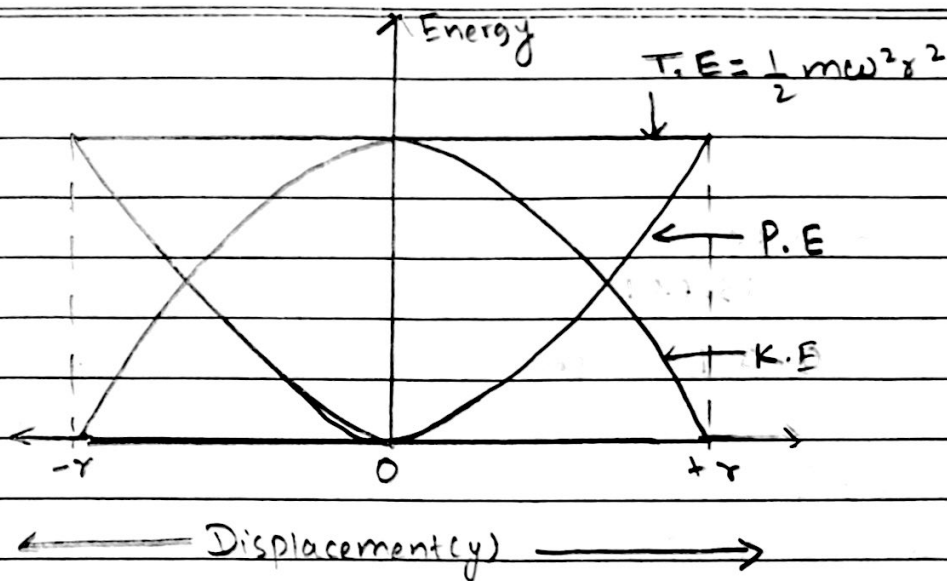
y	0	$\pm r$
P.E	0	$\frac{1}{2} m \omega^2 r^2$

Note

$$P.E_{\text{max}} = K.E_{\text{max}} \\ = T.E = \frac{1}{2} m \omega^2 r^2$$

$$\text{And. } K.E = \frac{1}{2} m \omega^2 (r^2 - y^2)$$

y	0	$\pm r$
K.E	$\frac{1}{2} m \omega^2 r^2$	0



8. A small mass of 0.2 kg is attached to one end of helical spring and produces an extension of 15 mm. The mass is now set into vertical oscillation of amplitude 10 mm. What is

- (i) The period of oscillation.
- (ii) The maximum K.E of the mass
- (iii) The P.E of the spring when the mass is 5 mm below the centre of oscillation? ($g = 9.8 \text{ m/s}^2$)

✓ Soln.

$$m = 0.2 \text{ kg}$$

$$e = 15 \text{ mm} = 15 \times 10^{-3} \text{ m}$$

$$r_{\text{max}} = 10 \text{ mm} = 10 \times 10^{-3} \text{ m}$$

(i) $T = ?$

Since,

$$F = ke$$

$$mg = ke$$

$$0.2 \times 9.8 = k \times 15 \times 10^{-3}$$

$$\therefore k = 1306.67$$

Then,

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$= 2\pi \sqrt{\frac{0.2}{1306.67}}$$

$$= 0.07 \text{ sec}$$

(ii) K.E_{max} = ?

$$K.E_{\text{max}} = \frac{1}{2} m \omega^2 r^2$$

$$= \frac{1}{2} \times 0.8 \times \left(\frac{2\pi}{T} \right)^2 \times r^2$$

$$= 0.26 \text{ kg m}^2/\text{s}^2$$

(iii) P.E = ? when $y = 5 \text{ mm} = 5 \times 10^{-3} \text{ m}$

$$P.E = \frac{1}{2} m \omega^2 (y^2)$$

$$= \frac{1}{2} m \left(\frac{2\pi}{T} \right)^2 \times y^2$$

$$= \frac{1}{2} \times 0.2 \times \left(\frac{2\pi}{T} \right)^2 \times$$

⑤ Calculate the period of oscillation of a simple pendulum of length 1.8 m with a bob of mass 2.2 kg. If bob of pendulum is pulled aside a horizontal distance of 20 cm & released. What will be K.E & Velocity of bob at lowest point?

$$T = ?$$

$$L = 1.8 \text{ m}$$

$$m = 2.2 \text{ kg}$$

$$r = 20 \text{ cm} = 0.2 \text{ m}$$

- ① $K.E = ?$ } at lowest point.
② $V = ?$ }

Now,

$$\begin{aligned} T &= 2\pi \sqrt{\frac{L}{g}} \\ &= 2\pi \sqrt{\frac{1.8}{9.8}} \\ &= 2.69 \text{ sec} \end{aligned}$$

$$\textcircled{1} \quad \cancel{K.E} \quad \omega = \frac{2\pi}{T} = \frac{2\pi}{2.69} = 2.33 \text{ rad/sec}$$

then,

$$K.E = \frac{1}{2} m \omega^2 (x^2 - y^2)$$

At lowest point, $y = 0$

$$\therefore K.E = \frac{1}{2} m \omega^2 x^2$$

$$= \frac{1}{2} \times 2.2 \times (2.33)^2 \times (0.2)^2$$

$$= 0.24 \text{ J}$$

$$\textcircled{2} \quad V = \omega \sqrt{x^2 - y^2}$$

At lowest point, $y = 0$

$$\therefore V = \omega x$$

$$V = 2.33 \times 0.2$$

$$\therefore V = 0.466 \text{ m/s}$$

• Angular SHM

→ The periodic motion in which ~~acceler~~ angular acceleration of a body is directly proportional to its angular displacement and ~~di~~ the angular acceleration is directed opposite to the angular displacement is called Angular SHM.

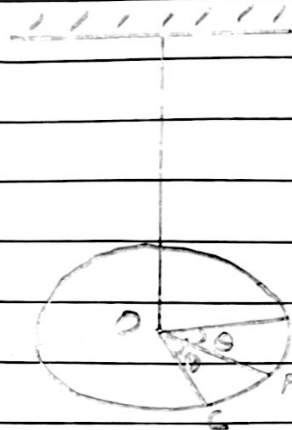


Fig - Torsional pendulum

Consider a metallic disc is suspended from a rigid support by a wire. When the disc deflects from reference line OA, then the wire gets twisted. The Twisted wire provides restoring torque to the disc. It is found that restoring torque is directly proportional to angle of deflection of the disc.

$$\text{i.e. } \tau \propto \theta$$

$$\therefore \tau = -k\theta \rightarrow (1)$$

Where k is Torsional Constant.

Since,

$$\tau = I\alpha \rightarrow (2)$$

From ① & ②

$$I \alpha = -K \theta$$

$$\therefore \alpha = -\frac{K}{I} \theta \rightarrow \textcircled{3}$$

This proves that the motion of torsional pendulum is angular SHM.

For angular SHM, we have

$$\alpha = -\omega^2 \theta \rightarrow \textcircled{4}$$

From ③ & ④.

$$\theta \times \frac{K}{I} = \omega^2 \theta$$

$$\text{or, } \omega^2 = \frac{K}{I}$$

$$\therefore \omega = \sqrt{\frac{K}{I}}$$

$$\text{or, } \frac{2\pi}{T} = \sqrt{\frac{K}{I}}$$

$$\therefore T = 2\pi \sqrt{\frac{I}{K}} \rightarrow \textcircled{5}$$

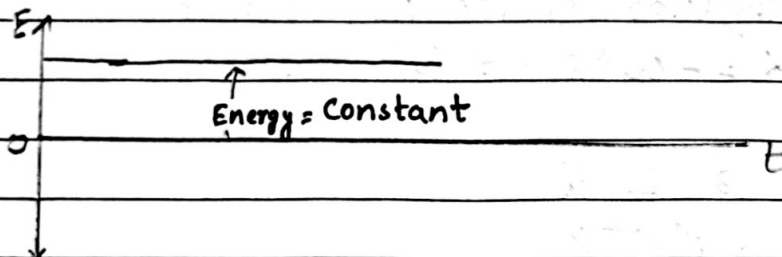
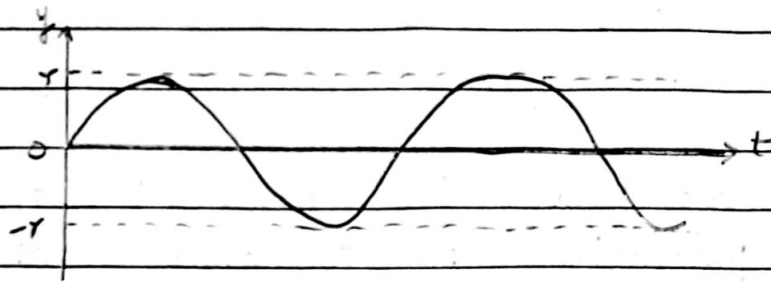
This gives time period of angular SHM.

> Types of Oscillation

⇒ ① Free Oscillation

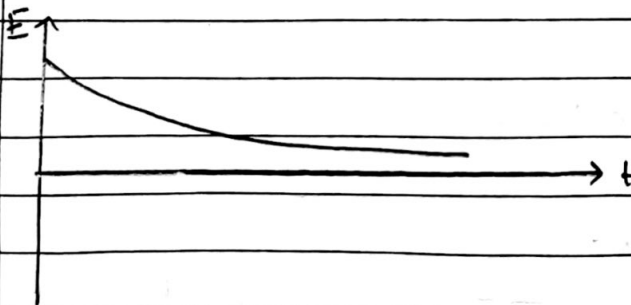
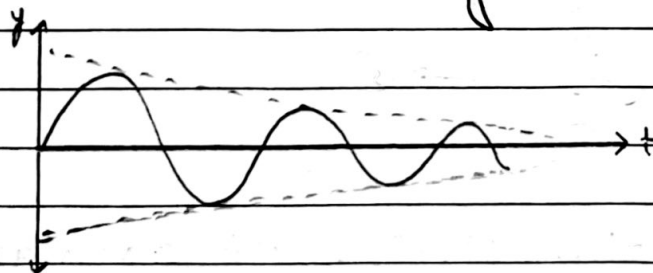
→ A particle ~~with~~ is said to be in free oscillation if its amplitude

and energy of oscillation remains same at all the time.



② Damped Oscillation

→ A body is said to be in damped oscillation if its amplitude and energy of oscillation decreases continuously with time.



⑪ Forced Oscillation

→ A body is said to be in Forced oscillation if its amplitude and Energy of oscillation remain constant under the action of external periodic force applied to it.

• Resonance

→ In Forced oscillation, if frequency of applied periodic force becomes equal to the natural frequency of oscillating body then the body oscillates with maximum amplitude. This phenomenon is called resonance.

Qn Why marching soldiers are ordered to break their steps while crossing a bridge?

→ If marching soldiers are not ordered to break their steps then, the frequency of marching may coincide with the natural frequency of bridge. If it happens then resonance occurs & the bridge oscillate with maximum amplitude which may cause the collapsing of bridge.

• Conceptual (short) Questions.

① Define simple harmonic motion.

↳ The back and forth motion of a particle in which restoring force acting on the particle is directly proportional to its displacement from the mean position and the restoring force is directed ~~proportional~~ towards mean position is called Simple harmonic motion.

② A simple harmonic motion is represented, in usual notation, by $y = a \sin(\omega t + \phi)$. Find its acceleration.

$$\hookrightarrow y = a \sin(\omega t + \phi)$$

$$\frac{dy}{dt} = a \frac{d \sin(\omega t + \phi)}{d(\omega t + \phi)} \times \frac{d\omega t}{dt} + \frac{d\phi}{dt}$$

$$v = a \cos(\omega t + \phi) \times \omega$$

$$v = a\omega \cos(\omega t + \phi)$$

$$\frac{dv}{dt} = \frac{d[a\omega \cos(\omega t + \phi)]}{dt}$$

$$= a\omega \frac{d \cos(\omega t + \phi)}{d(\omega t + \phi)} \times \frac{d\omega t}{dt}$$

$$= a\omega (-\sin(\omega t + \phi)) \times \omega$$

$$= -a\omega^2 \sin(\omega t + \phi)$$

$$\therefore \text{acceleration} = -a\omega^2 \sin(\omega t + \phi)$$

③ A SHM is represented as $y = a \cos(\omega t - \phi)$ in usual notation. Find its acceleration.

$$\hookrightarrow y = a \cos(\omega t - \phi)$$

$$\frac{dy}{dt} = a \frac{d \cos(\omega t - \phi)}{d(\omega t - \phi)} \times \frac{d(\omega t - \phi)}{dt}$$

$$V = a (-\sin(\omega t - \phi)) \times \omega$$

$$V = -a\omega \sin(\omega t - \phi)$$

$$\frac{dv}{dt} = -a\omega \frac{d \sin(\omega t - \phi)}{d(\omega t - \phi)} \times \omega$$

$$= -a\omega^2 \times \cos(\omega t - \phi)$$

- ④ sketch the displacement-time graph and velocity-time graph for a particle executing SHM. What is the phase difference between displacement and velocity?

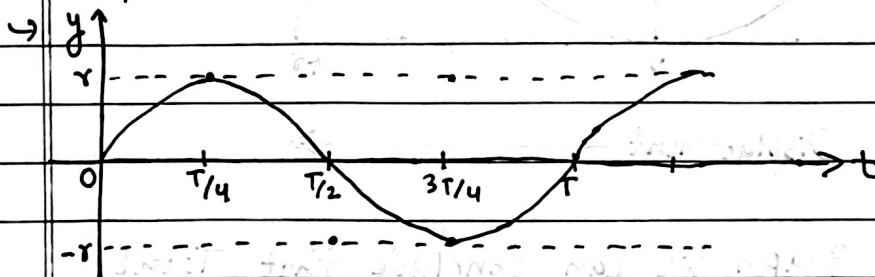


fig - ~~displacement~~ displacement-time graph.

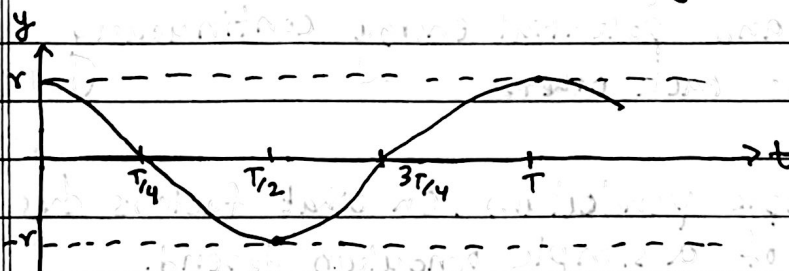
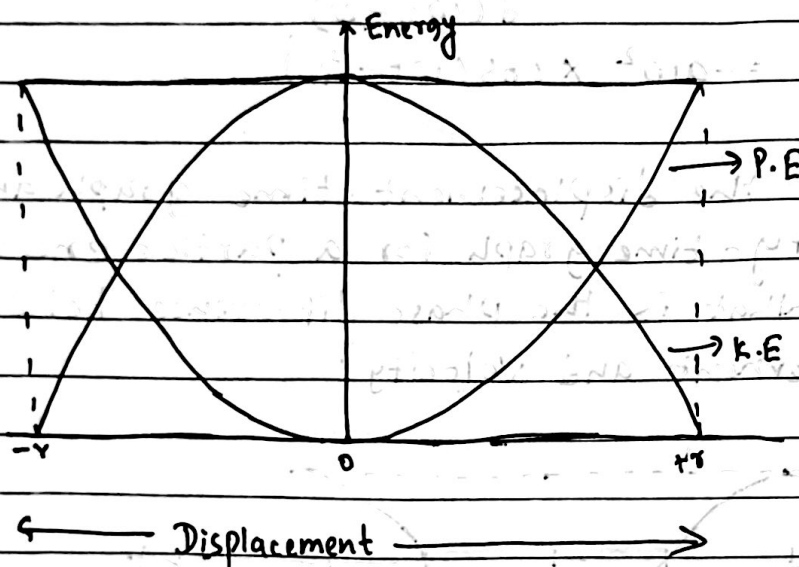


fig: Velocity-time graph

The phase difference between displacement and velocity is 90° or $\frac{\pi}{2}$ rad.

- ⑤ Sketch the graph showing the variation of potential energy and kinetic energy with displacement of particle executing SHM. What conclusion can you draw from the graph?

→



From the graph we can conclude that Total mechanical energy in SHM remains conserved as kinetic and potential energy continuously convert into each other.

- ⑥ Define simple pendulum. On what factors does the period of a simple pendulum depend.

→ A simple pendulum is a heavy particle suspended from a rigid support by an inextensible and weightless string which is free to oscillate in a vertical plane.

→ Period of a simple pendulum depends on:-

- i) Weight ($W = mg$)
- ii) Tension, T .

⑦ What do you understand by a second's pendulum? If it is taken to moon, will it gain or lose time? Why?

→ The simple pendulum having timeperiod 2 second is called second's pendulum.

→ If it is taken to moon, it will lose time.

As we know,

$$T \propto \frac{1}{\sqrt{g}}$$

$$g_m = \frac{g_e}{6}$$

$$T_m = 2\pi \sqrt{\frac{L}{g_m}} = 2\pi \sqrt{\frac{L}{g_e/6}} = \sqrt{6} \left(2\pi \sqrt{\frac{L}{g_e}} \right)$$

$$T_m = \sqrt{6} \cdot T_e$$

$$T_m = \sqrt{6} \times 2 \approx 4.90 \text{ s.}$$

$$T_m > T_e \Rightarrow \text{loses time}$$

⑧ If length of a simple pendulum increased by 4 times its original length, find the percentage change in its time-period.

→ let, $l_1 = x$

$$T_1 = 2\pi \sqrt{\frac{l_1}{g}} = 2\pi \sqrt{\frac{x}{g}}$$

$$l_2 = 4x$$

$$T_2 = 2\pi \sqrt{\frac{4x}{g}} = 2 \cdot 2\pi \sqrt{\frac{x}{g}}$$

$$T_2 = 2T_1$$

Now,

$$\% \text{ change} = \left(\frac{T_2 - T_1}{T_1} \right) \times 100\%$$

$$= \left(\frac{2T_1 - T_1}{T_1} \right) \times 100\%$$

$$= \left(\frac{T_1}{T_1} \right) \times 100\%$$

$$= 100\%$$

✓

⑨ Discuss the effect on time period of simple pendulum taken inside elevator when:

i) elevator is accelerating upward.

$$\begin{aligned} \rightarrow g_{\text{eff}} &= g + a \\ T' &= 2\pi \sqrt{\frac{L}{g_{\text{eff}}}} \\ &= 2\pi \sqrt{\frac{L}{g + a}} \end{aligned}$$

$T' < T \Rightarrow$ Time period decreases.

ii) elevator is in free-fall.

$$\rightarrow a = g.$$

$$g_{\text{eff}} = g - a = g - g = 0$$

$$T' = 2\pi \sqrt{\frac{L}{0}} = \infty$$

$T' = \infty \Rightarrow$ pendulum stops oscillating.

(10) What are the drawbacks of simple pendulum.

- ↳ The drawbacks of simple pendulum are:
- i) Real string has mass ($m_{\text{str}} \neq 0$).
 - ii) Bob is not a perfect point mass ($r_{\text{bob}} \neq 0$).
 - iii) Air resistance causes damping ($F_{\text{drag}} \neq 0$).
 - iv) Error at large angles ($\theta > 10^\circ \Rightarrow \sin \theta \neq \theta$)

• Exam pattern Questions

1c) The tip of tuning fork goes through 550 complete vibrations in 1 sec. Find the angular frequency and time period of the motion.

↳ Soln,

$$\cancel{f} \neq \cancel{550} f = 550 \text{ rev/s}$$

$$\omega = ?$$

$$T = ?$$

We know that,

$$\begin{aligned} \omega &= 2\pi f = 2 \times \pi \times 550 \\ &= 3455.75 \text{ rad/s} \end{aligned}$$

$$T = \frac{2\pi}{\omega}$$

$$T = \frac{2\pi}{3455.75}$$

$$= 1.81 \times 10^{-3} \text{ s}$$

$$= 1.81 \times 10^{-3} \text{ s}$$

$$\therefore T = 1.81 \times 10^{-3} \text{ s}$$

2.6) A particle executing linear SHM has a max. velocity of 40 cm s^{-1} and max. acceleration of 50 cm s^{-2} . Find its amplitude and period of oscillation.

→ Soln,

$$V_{\text{max}} = 40 \text{ cm s}^{-1}$$

$$a_{\text{max}} = 50 \text{ cm s}^{-2}$$

$$r = ?$$

$$T = ?$$

We know that,

$$V_{\text{max}} = \omega r$$

$$\text{or, } 40 = \omega r$$

$$\text{or, } r = \frac{40}{\omega} \rightarrow \textcircled{1} \Rightarrow \omega = \frac{40}{r}$$

$$a_{\text{max}} = \omega^2 r$$

$$\text{or, } 50 = \omega^2 r$$

$$\text{or, } 50 = \left(\frac{40}{r} \right)^2 \times r$$

$$\text{or, } 50 = \frac{1600}{r} \times r$$

$$\text{or, } r = \frac{1600}{50}$$

$$\therefore r = 32 \text{ cm} = 0.32 \text{ m}$$

Again,

$$\omega = \frac{40}{r} = 12.5 \text{ rad/s}$$

$$\bullet 32 \text{ cm}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{12.5} = 5.03 \text{ s}$$

3.c) A body of mass 2 kg suspended from a spring of negligible mass and is found to stretch the spring 0.1 m. what is its force constant and the time period?

↳ Soln,

$$m = 2 \text{ kg}$$

$$e = 0.1 \text{ m}$$

$$k = ?$$

$$T = ?$$

We know that,

$$F = ke$$

$$mg = k \times 0.1$$

$$\text{or, } 2 \times 10 = k \times 0.1$$

$$\text{or, } \frac{20}{0.1} = k$$

$$\therefore k = 200 \text{ N/m}$$

Again,

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$= 2\pi \sqrt{\frac{2}{200}}$$

$$= 2\pi \sqrt{\frac{1}{100}}$$

$$= \frac{2\pi}{10}$$

$$= \frac{\pi}{5}$$

$$= 0.62 \text{ s}$$

$$\therefore T = 0.62 \text{ s}$$

Q. A bob of mass 8 kg performs SHM of amplitude 30 cm. The restoring force is 60 N when max. displacement 30 cm. Calculate:

i) Time period.

ii) max. acceleration.

iii) K.E when displacement is 12 cm.

→ Soln,

$$m = 8 \text{ kg}$$

$$r = 30 \text{ cm} = 0.3 \text{ m}$$

$$F = 60 \text{ N}$$

$$i) F = k \cdot r$$

$$\text{or } 60 = k \times 0.3$$

$$\text{or } k = \frac{60}{0.3}$$

$$k = 200 \text{ N/m}$$

Now,

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$= 2\pi \sqrt{\frac{8}{200}}$$

$$= 2\pi \sqrt{\frac{1}{25}}$$

$$= \frac{2\pi}{5}$$

$$= 1.25 \text{ Sec}$$

$$ii) a_{\text{max}} = \omega^2 r$$

$$= \frac{k}{m} \times r$$

$$= \frac{200}{8} \times 0.3$$

$$= 7.5 \text{ m/s}^2$$

$$\therefore a_{\text{man}} = 7.5 \text{ m/s}^2$$

$$\text{iii) } K \cdot E = \frac{1}{2} K (x^2 - y^2)$$

$$= \frac{1}{2} \times 200 (0.3^2 - 0.12^2)$$

$$= 100 \times 0.0756$$

$$= 7.56 \text{ J}$$

b) If a uniform spring is cut in half, what is the force constant of each half? How would the frequency of half spring differ from that using the same mass and entire spring?

→ Soln,

$$K \propto \frac{1}{L} \rightarrow \textcircled{i}$$

$$K' \propto \frac{1}{L/2}$$

$$K' \propto \underline{2} \rightarrow \textcircled{ii}$$

$$K' = 2K$$

Here,

The force constant is doubled

Again,

frequency of half spring

$$f = \frac{1}{2\pi} \sqrt{\frac{K}{m}}$$

frequency of half spring.

$$f' = \frac{1}{2\pi} \sqrt{\frac{2K}{m}}$$

$$= \sqrt{2} f.$$

Hence,

The frequency of half spring increases by a factor of $\sqrt{2}$.

⑧ A tray of mass 12 kg is supported by two identical springs as shown in the figure. When the tray is pressed slightly and released, it executes SHM with a period of 1.5 s.

i) What is the force constant of ~~spring~~ each spring.

→ Soln,

$$m = 12 \text{ kg}$$

$$t = 1.5 \text{ s}$$

~~k_{eq}~~ = ~~k~~ For ^{Parallel} ~~series~~ combination,
~~k_{eq}~~ $\Rightarrow k_{eq} = k + k = 2k$

$$\text{or, } \frac{1}{k} = \frac{2}{k_{eq}}$$

$$\text{or, } k = \frac{k_{eq}}{2} \Rightarrow \text{~~k = 2k_{eq}}~~$$

Now,

$$t = 2\pi \sqrt{\frac{m}{k_{eq}}}$$

$$\text{or, } 1.5 = 2\pi \sqrt{\frac{12}{k}}$$

Squaring on both side,

$$\frac{9}{4} \times \frac{1}{4\pi^2} = \frac{12.6}{2k}$$

or, $K_s = \frac{947.45}{9}$

$\therefore K_s = 105.27 \text{ N/m}$

ii) When a block of mass 'M' is placed on the tray, the period of SHM changes to 3.0s. calculate M.

→ Soln,

~~2π~~ $T = 2\pi \sqrt{\frac{M+12}{K_s}}$

or, $3 = 2\pi \sqrt{\frac{M+12}{K_s}}$

or, $(0.47)^2 = \frac{M+12}{K_s} = \frac{M+12}{105.27}$

or, $23.9986 + 12 = M$

or, $M = 36 \text{ kg}$

6. a) A body is executing SHM with Amplitude 'A' and period of 'T'. In terms of T, find;

i) time taken by particle to reach a displacement equal to half the amplitude starting from the mean position.

→ ~~Y~~ $T = T_s$

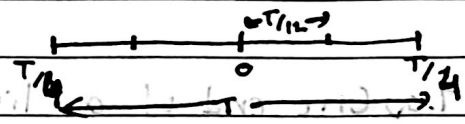
$y = A \sin(\omega t)$

$y = A \sin(\omega t) = A \sin(\omega t)$

or, $\frac{A}{2} = A \sin(\omega t)$

or, $\sin(\omega t) = \frac{1}{2}$

or, $\omega t = 30 = \frac{\pi}{6}$



$$\text{or, } \left(\frac{2\pi}{T}\right)t = \frac{\pi}{6}$$

$$\text{or, } t = \frac{\pi}{6} \times \frac{T}{2\pi} = \frac{T}{12}$$

ii) Time taken by particle to reach a displacement equal to half the amplitude starting from the extreme position.

→ Soln,

$$\frac{T}{12} + t = \frac{T}{4}$$

$$\text{or, } t = \frac{T}{4} - \frac{T}{12} = \frac{2}{12} T = \frac{1}{6} T$$

b) The displacement of an oscillating object as function of time is shown in fig. Calculate its time period, frequency and amplitude.

→ Soln,

$$r = 0.20 \text{ m}$$

$$t = 2.0 \text{ s}$$

$$f = \frac{1}{t} = \frac{1}{2} = 0.5 \text{ Hz}$$

7. a) One end of a light spring constant 18 N/m is attached to a rigid support. A mass of 0.15 kg is attached to the other end and mass is pulled such that extension of spring is 4 cm . The student releases the mass such that it performs horizontal SHM. Calculate

i) Max. acceleration and time period of SHM

✓ Soln,

$$K = 18 \text{ N/m}$$

$$m = 0.15 \text{ kg}$$

$$x_0 = 4 \text{ cm} = 0.04 \text{ m}$$

$$\omega = \sqrt{\frac{K}{m}} = \sqrt{\frac{18}{0.15}} = 10.95$$

Now,

$$a_{\text{max}} = \omega^2 x$$

$$= (10.95)^2 \times 0.04$$

$$= 4.8 \text{ m/s}^2$$

ii) Max. kinetic energy.

$$\hookrightarrow K_{\text{max}} = \frac{1}{2} K x^2$$

$$= \frac{1}{2} \times 18 \times (0.04)^2$$

$$= 0.0144 \text{ J}$$

$$T = 2\pi \sqrt{\frac{m}{K}} = 2\pi \sqrt{\frac{0.15}{18}}$$

$$= 0.57 \text{ s}$$